EXP 18

```
clc;
clear;
close all;

% Eye Diagram Analysis under Inter-Symbol Interference (ISI)
% No Communications Toolbox required

% Step 1: Parameters
N = 1000;            % Number of bits
sps = 20;            % Samples per symbol
SNR_dB = 20;         % Signal-to-noise ratio

% Step 2: Generate Random Binary Data
data = randi([0 1], 1, N);

% Step 3: BPSK Modulation
symbols = 2*data - 1;

% Step 4: Upsampling
tx = zeros(1, N*sps);
tx(1:sps:end) = symbols;
```

```
% Step 5: Create Channel with ISI
% Current symbol is affected by neighboring symbols
h = [0.25 1 0.45];

% Apply channel response
rx_isi = conv(tx, h, 'same');

% Step 6: Add AWGN
signal_power = mean(rx_isi.^2);
noise_power = signal_power / (10^(SNR_dB/10));
rx = rx_isi + sqrt(noise_power)*randn(size(rx_isi));

% Step 7: Plot Transmitted Signal
figure;
plot((0:10*sps-1)/sps, tx(1:10*sps), 'b');
grid on;
xlabel('Time (Symbol Periods)');
ylabel('Amplitude');
title('Transmitted BPSK Signal');

% Step 8: Eye Diagram without ISI
figure;
hold on;
```

```
for k = 1:2:N-2
    idx = (k-1)*sps + 1;
    if idx + 2*sps - 1 <= length(tx)
        segment = tx(idx:idx+2*sps-1);
        plot((0:2*sps-1)/sps, segment, 'b');
    end
end

grid on;
xlabel('Time (Symbol Periods)');
ylabel('Amplitude');
title('Eye Diagram without ISI');
xlim([0 2]);
hold off;

% Step 9: Eye Diagram with ISI and Noise
figure;
hold on;

for k = 1:2:N-2
    idx = (k-1)*sps + 1;
    if idx + 2*sps - 1 <= length(rx)
        segment = rx(idx:idx+2*sps-1);
```

```
        plot((0:2*sps-1)/sps, segment, 'r');
    end
end

grid on;
xlabel('Time (Symbol Periods)');
ylabel('Amplitude');
title('Eye Diagram with ISI and AWGN');
xlim([0 2]);
hold off;

% Step 10: Evaluate Eye Diagram Parameters
% Sample at symbol centers
sample_idx = 1:sps:length(rx);
samples = rx(sample_idx);

% Estimate eye height from symbol-center samples
level1 = samples(data == 0);
level2 = samples(data == 1);

eye_height = mean(level2) - mean(level1);
noise_spread = std(samples);

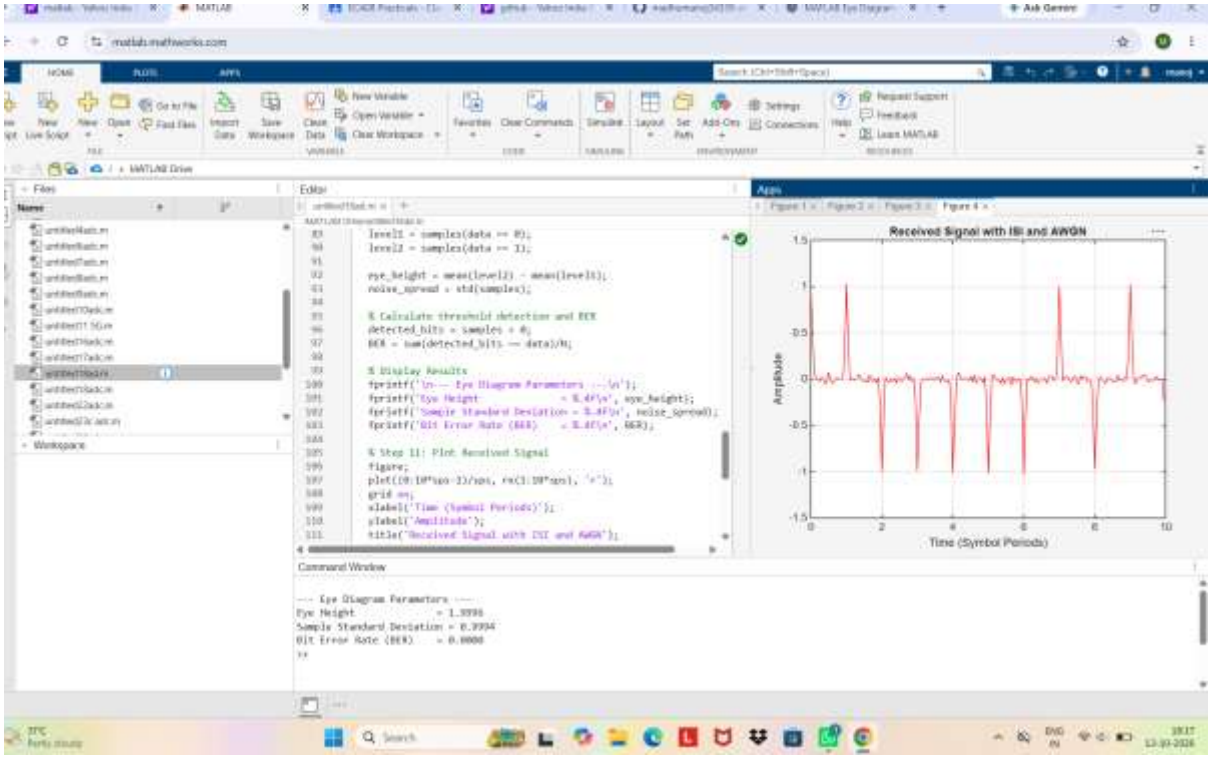
% Calculate threshold detection and BER
```

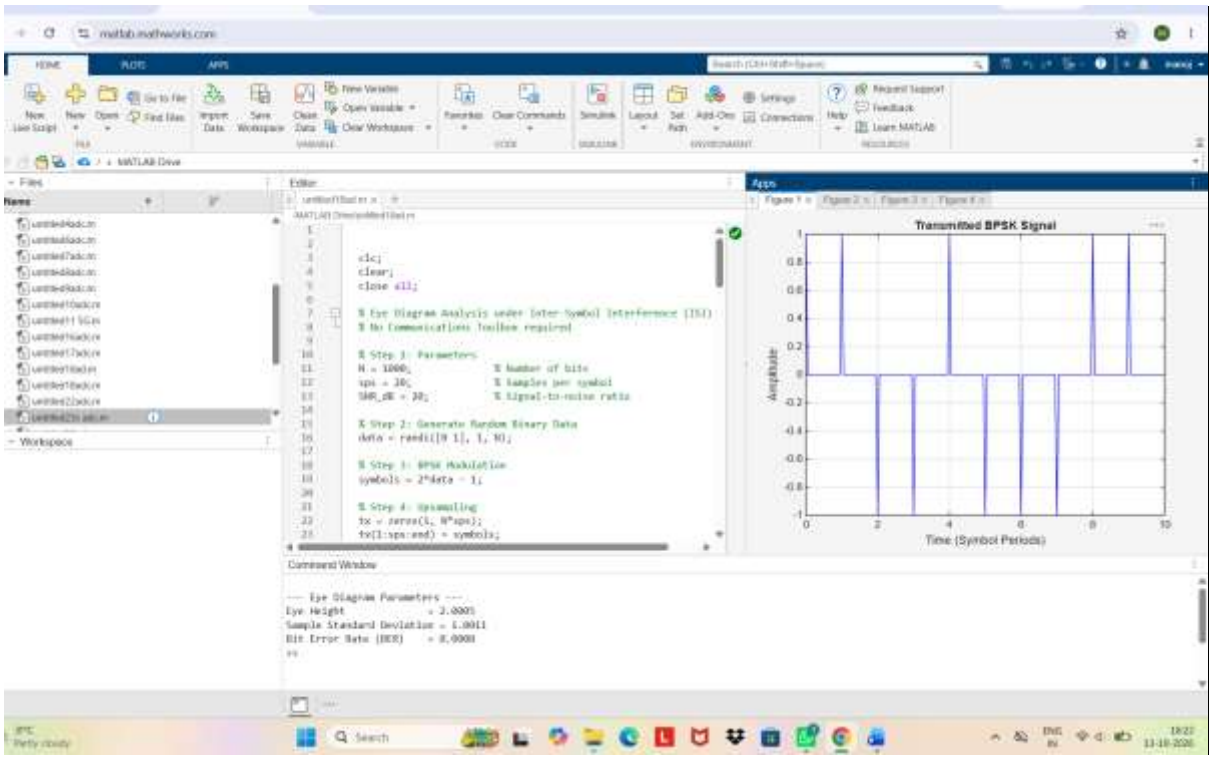
```
detected_bits = samples > 0;
BER = sum(detected_bits ~= data)/N;

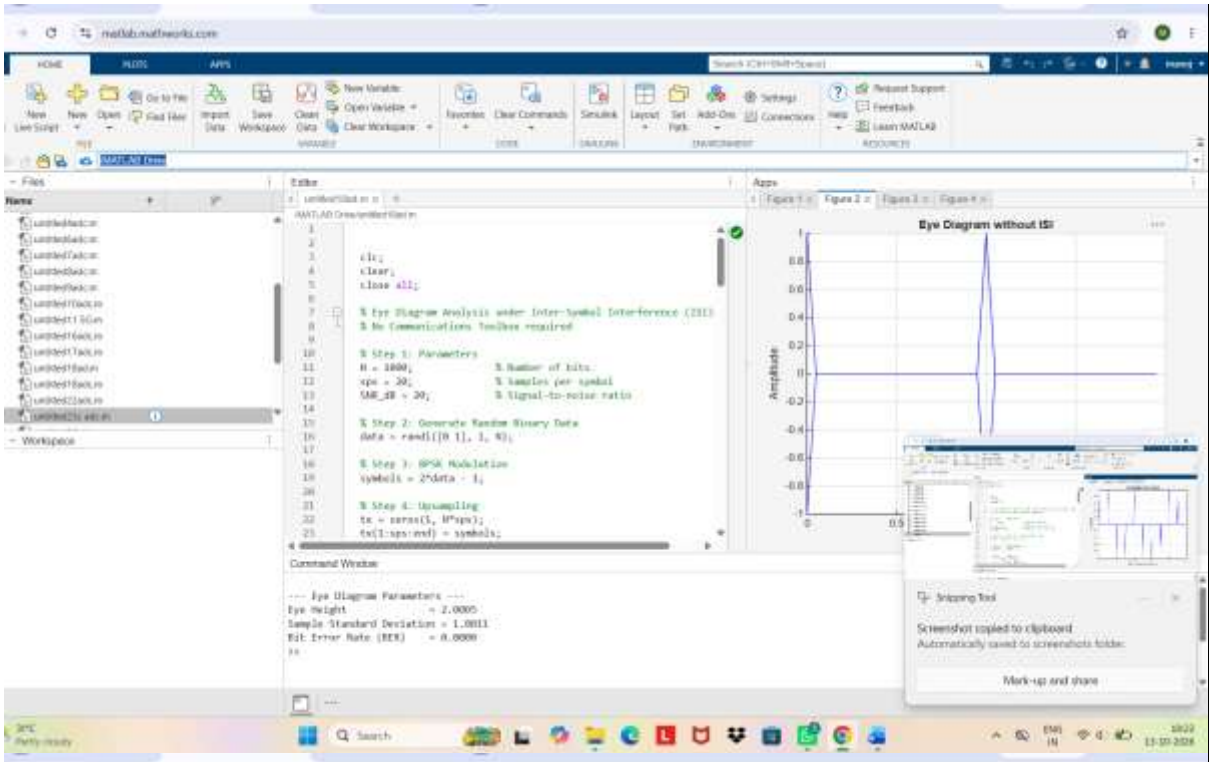
% Display Results
fprintf('\n--- Eye Diagram Parameters ---\n');
fprintf('Eye Height             = %.4f\n', eye_height);
fprintf('Sample Standard Deviation = %.4f\n', noise_spread);
fprintf('Bit Error Rate (BER)    = %.4f\n', BER);

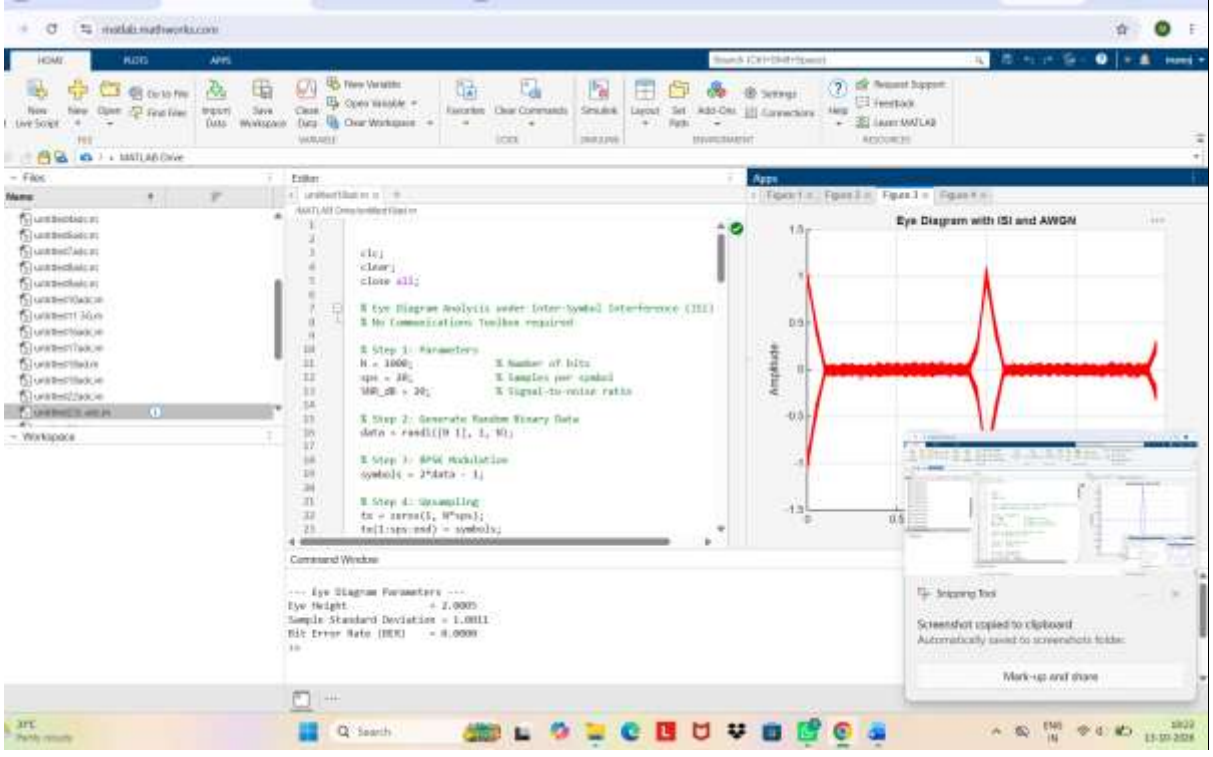
% Step 11: Plot Received Signal
figure;
plot((0:10*sps-1)/sps, rx(1:10*sps), 'r');
grid on;
xlabel('Time (Symbol Periods)');
ylabel('Amplitude');
```

title('Received Signal with ISI and AWGN');






Eye Diagram without ISI
Amplitude
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Mark-up and share


Eye Diagram with ISI and AWGN
Amplitude
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